Global Explainer:
Dual-Paper Project for Physics-Based Lesion Phenotyping

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March 2026

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1 Project Overview

This document explains the full body of work behind two journal-ready manuscripts on automated, unsupervised lesion segmentation using Navier-Stokes active contours with per-image physics parameter optimization.

The core pipeline treats lesion segmentation as a black-box optimization problem: for each of 174 high-resolution soybean leaf images, a stochastic search finds the 7-parameter vector $\theta = (\mu, \lambda, d, \alpha, \beta, \gamma, E_{\text{thr}})$ that maximizes a no-ground-truth heuristic objective combining gradient alignment, chromatic separability, and topological constraints. The full optimizer costs approximately 260 seconds per leaf.

This expensive but effective oracle creates a natural two-paper structure: one paper about *building and analyzing the oracle itself*, and one paper about *approximating it efficiently for deployment*.

2 Paper Split Rationale

2.1 Why Two Papers?

The repository contains evidence for two distinct scientific contributions that are stronger when presented separately than when collapsed together:

1. **Paper 1 (Optimizer/Oracle):** The design, analysis, and characterization of the physics-based optimizer. This includes the objective function, search strategy, ablations, trace-based landscape analysis, and parameter-cloud geometry. The contribution is methodological: a new way to do unsupervised physics-based segmentation.

2. **Paper 2 (Deployment/Runtime):** The operational problem of approximating the expensive oracle at deployment time. This includes selective semi-amortized routing, full-leaf vs patchwise comparison, calibration, runtime-quality tradeoffs, and failure analysis. The contribution is practical: explicit operating points for practitioners with limited compute.

2.2 What Belongs Where

- **Paper 1 only:** Objective function design, search strategy, ablation studies, trace-based surface reconstruction, parameter-manifold geometry (PCA, t-SNE, UMAP, solid 3D), distillation as secondary consequence.

- **Paper 2 only:** Selective routing policy, runtime-quality Pareto frontier, calibration/risk-coverage, aggregation method comparison, oracle gap quantification, failure taxonomy, cross-collection robustness.

- **Shared (minimal):** Refinement budget table (different framing in each), per-parameter difficulty (optimizer interpretation in P1 vs routing interpretation in P2).

3 Paper 1: Gradient-Free Stochastic Optimization of Navier-Stokes Active Contours

3.1 Central Question

Can a gradient-free stochastic optimizer reliably find per-image physics parameters for Navier-Stokes active contour segmentation without ground truth?

3.2 Key Results

- 100% convergence on all 174 images (mean score 21.10, mean runtime 258s)
- Objective ablations: topology term contributes +11.12 points, gradient alignment +5.09 points
- Search ablations: random exploration reaches ∼90% of final score; refinement adds ∼10%
- Trace-based surface analysis on 9 representative leaves reveals score-tier-dependent landscape geometry: sharp peaks for high-score leaves, flat surfaces for low-score leaves
- Parameter-cloud geometry: 2 PCs capture 46.5% of variance, centroid shift 1.616 standardized units from global to local optimization
- k-NN agreement 0.534 (vs 0.33 random baseline) shows parameter shift direction predicts optimization success
- UNet student: mean Dice 0.598 (median 0.682) across 174 images
- Local re-optimization reduces count error by 35%

3.3 Manuscript Details

18-page IEEEtran journal paper. 19 main-text figures, 3 tables, 27 verified references. Three appendices with additional optimizer diagnostics, trace plots, and geometry supplementary figures.

4 Paper 2: Black-Box Selective Semi-Amortized Inference

4.1 Central Question

How should compute be allocated between direct prediction, short refinement, and full optimization for deployment?

4.2 Key Results

- On 35-leaf matched split: patchwise mean aggregation achieves nMAE 0.906 vs full-leaf 1.154 (21.5% improvement, $p = 0.003$)
- Oracle gap: 0.327 nMAE (irreducible mismatch between full-leaf and aggregated patch oracles)
- Selective routing operating points:
  - Quality-favoring: 0.556 nMAE at 74.34s (3.50× speedup)
  - Balanced: 0.577 nMAE at 60.48s (4.31× speedup)
  - Max-speedup: 0.819 nMAE at 55.77s (4.67× speedup)
- Calibration directionally correct (monotonic uncertainty-error relationship)
- Failure taxonomy: Both good 29%, FL only 20%, Patch only 20%, Both bad 31%
- Cross-collection robustness on 28 external leaves from 18.7 collection

4.3 Manuscript Details

20-page article-class paper. 9 multi-panel main-text figures, 13 tables, 19 verified references. 5 appendix figure families.

5 Experimental Infrastructure

5.1 Data

- **Primary:** 174 soybean leaf images, 16-bit TIFF, ∼3900×1340 px
- **External:** 78 images from collection 18.7 (cross-domain evaluation)
- **Patches:** 224×224 px at stride 112, ∼275 patches/leaf (47,916 total)

5.2 Optimizer

Random-restart stochastic hill climbing. 64 random candidates, top-4 refined with 12 steps each, downscale $s = 0.75$. 7 physics parameters: viscosity μ, elastic coupling λ, diffusion rate d, tension α, rigidity/balloon β, inflation γ, energy threshold E_{thr}.

5.3 Student Models

- UNet: patch-based (512×512), 3-level encoder-decoder, BCE+Dice loss, 40 epochs
- Feature extraction: DINOv2 (384D CLS token), ResNet-18 (512D avg-pool), Handcrafted (45D color+texture+shape)
- Regression heads: Ridge ($\alpha = 1.0$), MLP (2 hidden layers), Ensemble (5 models with std uncertainty)

6 Key Figure Guide

6.1 Paper 1 Figures

- Figs 1–3: Optimizer status, runtime, score histograms → `figures/exp_v1/optimizer_*.png`
- Fig 4: Threshold sweep → `figures/exp_v1/threshold_sweep_plot.png`
- Figs 5–6: Teacher-student montage and disagreement → `figures/exp_v1/montage_*.png`
- Figs 7–8: Parameter-morphology scatter → `figures/exp_v1/scatter_*.png`
- Figs 9–11: Trace-based objective surfaces (3 score tiers) → `figures/9leaf/surface_*.png`
- Fig 12: Convergence trajectory → `figures/9leaf/traj_*.png`
- Figs 13–15: PCA, mean delta, covariance ellipsoids → `figures/manifold/*.png` and `figures/solid3d/*.png`

6.2 Paper 2 Figures

- fig:setup_main: Pipeline + main comparison + oracle gap → `figures/B_fig01-05*.pdf`
- fig:frontierB: Runtime-quality Pareto → `figures/B_fig04,08,19,38*.pdf`
- fig:calibrationB: Risk-coverage + calibration → `figures/B_fig09,10,27,28*.pdf`
- fig:failureB: Failure taxonomy + cross-collection → `figures/B_fig18,35,36*.pdf`

7 Strongest Evidence

7.1 Paper 1

1. **Ablation tables** (Tables I–II): Systematic justification of each objective term and search strategy choice
2. **Trace-based surfaces** (Figs 9–11): Direct visualization of the true objective landscape explored by the optimizer
3. **PCA manifold** (Fig 13, centroid shift 1.616 std units): Demonstrates structured, non-random parameter organization

7.2 Paper 2

1. **Matched comparison** (Tab mainB, $p = 0.003$): Rigorous statistical evidence for patch-wise advantage
2. **Pareto frontier** (Fig frontierB): Multiple explicit operating points, not a single threshold
3. **Oracle gap** (0.327 nMAE): Honest quantification of fundamental resolution mismatch

8 What Was Reused vs New

Almost all evidence was reused from existing experimental outputs. The repository contained extensive outputs from:

- Production optimizer run (174 leaves)
- Objective and search ablations (30-image subset, 11 variants)
- 9-leaf representative trace study
- Parameter manifold geometry study (150 patches, 15 leaves)
- Integrated full-vs-patch comparison (35-leaf matched split)
- Selective semi-amortized routing study (174 leaves)
- Patchwise selective study (1,059+ patches)
- Teacher-student UNet distillation

Only lightweight synthesis assets were new:

- Runtime frontier plot (from `dual_paper_build_codex/gapfill/`)
- Risk overlay plot (from same)
- Three TODO placeholders in Paper 1 resolved with factual text

No heavy optimization reruns or new model training was performed.

9 Provenance

Detailed inventories are in `notes/`:

- `repository_audit.txt` âĂŤ Full directory-level audit
- `manuscript_line_inventory.csv` âĂŤ All manuscript versions tracked
- `output_family_inventory.csv` âĂŤ All experimental output families
- `dual_paper_evidence_matrix.csv` âĂŤ Claim-to-evidence mapping
- `dual_paper_gap_analysis.txt` âĂŤ What was weak or missing
- `dual_paper_distinction.txt` âĂŤ How the two papers differ
- `reference_master_audit.csv` âĂŤ All references verified
- `reference_quarantine.txt` âĂŤ Problematic references flagged
- `figure_and_table_guide.csv` âĂŤ Per-figure interpretation
- `claim_to_output_map.csv` âĂŤ Claims mapped to evidence